HealthConnect – System Requirements Specification

3. Requirements

3.1 Functional Requirements

Req#	Requirement	Description	Comments	Rate	Reviewed / Approved
FR_01	Patient Registration Management	Handles full patient registration including personal, guardian and medical details in a structured digital format.	Eliminates paperwork and reduces human error.	High	D.M.Dajlani / V.Beqaraj / D.Filja
FR_02	Appointment Scheduling	Supports booking, editing, and canceling appointments while checking doctor availability in real-time.	Improves hospital efficiency and reduces waiting time.	High	D.M.Dajlani / V.Beqaraj / D.Filja
FR_03	Doctor Assignment	Automatically or manually assigns doctors to patients based on specialization and workload.	Ensures balanced workload and better service quality.	High	D.M.Dajlani / V.Beqaraj / D.Filja
FR_04	Medical Records	Maintains detailed and continuously	Provides accurate and complete	High	D.M.Dajlani / V.Beqaraj

	Management	updated patient medical records, including diagnosis, treatments, and notes.	patient history.		/ D.Filja
FR_05	Bed Management	Tracks bed occupancy and availability in real-time, allowing efficient patient placement.	Prevents conflicts and improves resource utilization.	High	D.M.Dajlani / V.Beqaraj / D.Filja
FR_06	Discharge Management	Generates comprehensive discharge summaries with diagnosis, treatments, and follow-up instructions.	Ensures proper patient guidance after leaving hospital.	High	D.M.Dajlani / V.Beqaraj / D.Filja
FR_07	Patient Search System	Allows fast retrieval of patient data using multiple filters such as ID, name, or phone number.	Improves speed and avoids duplication.	High	D.M.Dajlani / V.Beqaraj / D.Filja
FR_08	User Authentication	Provides secure login using credentials and authentication mechanisms.	Protects sensitive hospital data.	High	D.M.Dajlani / V.Beqaraj / D.Filja
FR_09	Role-Based	Ensures that users can only	Maintains privacy and	High	D.M.Dajlani / V.Beqaraj

	Access Control	access features relevant to their roles.	data security.		/ D.Filja
FR_10	Reporting System	Generates detailed analytical reports about hospital operations and performance.	Supports management decisions.	Medium	D.M.Dajlani / V.Beqaraj / D.Filja
FR_11	Notifications and Alerts	Provides real-time alerts for appointments, emergencies, and updates.	Enhances responsiveness.	Medium	D.M.Dajlani / V.Beqaraj / D.Filja
FR_12	Data Backup and Recovery	Automatically backs up data and enables recovery in case of failure.	Prevents data loss.	High	D.M.Dajlani / V.Beqaraj / D.Filja
FR_13	Audit Logs	Records all user actions and system activities for monitoring and auditing.	Improves transparency and accountability.	Medium	D.M.Dajlani / V.Beqaraj / D.Filja
FR_14	Patient History Tracking	Maintains long-term records of all visits and treatments.	Supports accurate diagnosis.	High	D.M.Dajlani / V.Beqaraj / D.Filja
FR_15	Emergency Case Handling	Allows prioritization and fast handling of emergency patients.	Improves critical care response.	High	D.M.Dajlani / V.Beqaraj / D.Filja

FR_16	Prescription Management	Enables digital prescription creation and management.	Reduces medication errors.	High	D.M.Dajlani / V.Beqaraj / D.Filja
FR_17	Staff Management	Allows administrators to manage user accounts, roles, and permissions.	Improves administrative efficiency.	Medium	D.M.Dajlani / V.Beqaraj / D.Filja
FR_18	Dashboard Monitoring	Provides a real-time overview of system status and hospital activity.	Supports monitoring and decision-making.	Medium	D.M.Dajlani / V.Beqaraj / D.Filja

3.2 Non-Functional Requirements

Non-functional requirements define how the system operates and ensure quality attributes such as performance, security, usability, and reliability are achieved.

3.2.1 Product Requirements

Security Requirements

The system must ensure a high level of security due to the sensitivity of medical data.
• All patient data must be encrypted both at rest and in transit.
• Strong authentication mechanisms must be implemented.
• Role-based access control must restrict unauthorized access.
• Regular security audits and vulnerability testing must be performed.
• Secure backup systems must be implemented to protect data.
• The system must comply with healthcare data protection laws.

Performance Requirements

The system must perform efficiently under different conditions.
• System response time must be under 2 seconds for most operations.
• The system must support multiple concurrent users without degradation.
• Data retrieval must remain fast even with large datasets.
• The system must maintain performance during peak usage hours.

Reliability Requirements

The system must be reliable and continuously available.
• The system must operate 24/7 with minimal downtime.
• Automatic backups must occur regularly.
• The system must include failover mechanisms.
• A disaster recovery plan must be in place.

Usability Requirements

The system must be easy to use for all hospital staff.
• The interface must be simple and intuitive.
• Users must be able to learn the system quickly.
• The system must provide clear error messages.
• The design must reduce the likelihood of user mistakes.

Scalability Requirements

The system must be able to grow with the hospital needs.
• It must support increasing numbers of users and data.
• It must allow expansion to additional hospital departments.
• It must handle future upgrades efficiently.

Maintainability Requirements

The system must be easy to maintain and update.
• Code should be modular and well-structured.
• Updates and bug fixes must be easy to implement.
• Documentation must be clear and complete.

Compatibility Requirements

The system must work across multiple environments.
• It must be compatible with modern web browsers.
• It must function on different devices (PCs, tablets).
• It must integrate with existing hospital systems.

3.2.2 Organizational Requirements

• Training programs must be provided for all staff.
• Detailed user manuals must be available.
• Technical support must be accessible.
• System updates must follow structured procedures.

3.2.3 Environmental Requirements

• The system must use modern technologies (Java, HTML, CSS, JavaScript).
• It must run on secure servers.
• Network infrastructure must be protected by firewalls.
• Hardware must support system requirements.

3.2.4 External Requirements

• The system must integrate with other hospital systems.
• It must comply with legal and ethical standards.
• API communication must be secure.
• Patient privacy must be maintained at all times.

3.2.5 Availability Requirements

• The system must be available at least 99.9% of the time.
• Planned maintenance must be scheduled during low-usage hours.
• System downtime must be minimized and communicated in advance.

3.2.6 Data Integrity Requirements

• The system must ensure accuracy and consistency of all stored data.
• Data validation must be applied during input processes.
• No data should be lost during system operations or failures.

3.2.7 Logging and Monitoring Requirements

• The system must continuously monitor performance and usage.
• All errors and system issues must be logged automatically.
• Administrators must be notified in case of critical failures.

3.2.8 Response and Recovery Requirements

- The system must recover from failures within a short time (e.g., less than 5 minutes).
- Backup restoration must be tested regularly.
- System errors must not affect core functionalities.

3.2.9 User Experience Requirements

- The system must provide a consistent interface across all modules.
- Navigation must be simple and require minimal clicks.
- Important actions must be clearly highlighted.

3.2.10 Interoperability Requirements

- The system must support data exchange with external healthcare systems.
- Standard data formats (e.g., JSON, XML) must be supported.
- APIs must be well-documented and easy to use.

3.2.11 Audit and Compliance Requirements

- The system must maintain compliance with healthcare regulations.
- All user actions must be traceable.
- Audit reports must be generated when required.